Shahzaib Khan

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Education

Ph.D., **University of Washington** *June 2026*

- Civil and Environmental Engineering | Minor in Data Science Option
- Dissertation title: *From Sources to Sinks: Advancing Surface Water Management Through Satellite Remote Sensing*. Link to Defense PPT Slides.
- **# Publications:** Authored/co-authored 14 peer-reviewed publications (published or under review)

M.S., **University of Washington** *December 2022*

- Civil and Environmental Engineering
- Thesis title: *Understanding volume estimation uncertainty of lakes and wetlands using satellites and citizen science.* Related paper.

B.Tech (with Honors), **Indian Institute of Technology Gandhinagar (IIT-GN)** *July 2021*

- Department of Civil Engineering

Awards and Honors

- **Ronald & Mary Nece Award** (*Link*)**:** A prestigious honor awarded to top doctoral students in the University of Washington Department of Civil & Environmental Engineering for outstanding academic achievement *2026*
- **AWRA Washington State Section Fellowship** (*Link*)**:** A prestigious award awarded to top graduate students in Washington for excellence and leadership in interdisciplinary water resources research. *2023*
- **Appreciation Award:** From *Bangladesh Water Development Board* for training on utilizing SWOT satellite data for water resource management, *News Link* *2023*
- **Appreciation Award** (*Link*)**:** From *NASA* for training early adopters during the *SWOT Early Adopter Virtual Hackathon* at the *University of Washington* *2022*
- **IIT Gandhinagar Dean's List** (*Link*)**:** A prestigious academic honor awarded to students who maintain a GPA of 8.5/10.0 or higher while carrying a full-time course load. *2018-2021*

Peer-Reviewed Publications

1. M. Sharma, F. Hossain, S. Suresh, T. Pavelsky, S. Minocha, and **S. Khan** (2026). How Well Can the Surface Water and Ocean Topography (SWOT) Satellite Mission Observe Irrigation Canals of the Asian Green Revolution? *Geophysical Research Letters* (In Review)
2. **S. Khan,** Y. O. Wang, G. Darkwah, F. Hossain (2026). When the Cloud Warms the River: A Call for Monitoring Data Center Impacts in the Age of AI, *AGU Advances* (In Revision).
3. **S. Khan,** F. Hossain, M. Ahamed, and K. Islam (2026). Satellite Data Rendered Irrigation using Penman-Monteith and SEBAL – sDRIPS for Surface Water Irrigation Optimization, *Hydrology and Earth System Sciences (HESS),* https://doi.org/10.5194/hess-30-3675-2026.
4. **S. Khan,** Z. N. Hossain, S. Suresh, F. Hossain (2026). The Untapped Hydropower Potential of World's Cities, *Earth's Future*. https://doi.org/10.1029/2025EF007873.
5. **S. Khan** and F. Hossain (2026). (sDRIPS): A Cloud-Based, Open-Source Python Package for Satellite-Informed Surface Water Irrigation Optimization. *Digital Water* (In Press).
6. **S. Khan,** F. Hossain, et al. (2024). A Network Design Approach for Citizen Science-Satellite Monitoring of Surface Water Volume Changes in Bangladesh, *Environmental Modeling and Software*. https://doi.org/10.1016/j.envsoft.2023.105919.
7. A. M. Gómez, S. Biancamaria, T. Pavelsky, K. Nielsen, G. Parkins, M. Lane, **S. Khan**, F. Hossain, R. Bhattarai, S. Ghafoor, J.F. Crétaux, C. Yanez, N. Picot (2024). Evaluation using In-Situ Observations from National Governments and Citizen Scientists Suggests Nadir Altimeters can Accurately Measure Water Levels Changes Regardless of Lake Area. *GIScience and Remote Sensing*. https://doi.org/10.1080/15481603.2025.2543521.
8. G. Darkwah, F. Hossain, V. Tchervenski, G. Holtgrieve, C. Seaton, D. Graves, S. Minocha, P. Das, **S. Khan**, S. Suresh (2024) Reconstruction of the Hydro-Thermal History of Regulated River Networks Using Satellite Remote Sensing and Data-driven Techniques, *Earth's Future, vol 12(10)*, https://doi.org/10.1029/2024EF004815.

9. S. Minocha, F. Hossain, P. Das, S. Suresh, **S. Khan,** G. Darkwah, K. Andreadis, H. Lee, G. Holt, S. Galelli (2023). Reservoir Assessment Tool: A scalable and easy-to-apply python based software architecture to empower the global water community, *Geoscientific Model Development*, https://doi.org/10.5194/gmd-2023-130.
10. S. Suresh, F. Hossain, S. Minocha, P. Das, **S. Khan**, H. Lee, K. Andreadis and Perry Oddo (2023). Satellite-based Tracking of Reservoir Operations for Flood Management during the 2018 Extreme Weather Event in Kerala, India, *Remote Sensing of Environment, vol. 307*, https://doi.org/10.1016/j.rse.2024.114149.
11. S. Minocha, Pei-Hsin Pei, **S. Khan**, and F. Hossain (2023). Factors influencing Lake Surface Temperature for Reservoirs of the Columbia River Basin**,** *Northwest Science, vol. 97(4)*. https://doi.org/10.3955/046.097.0403
12. **S. Khan,** F. Hossain, et al. (2022) Understanding Volume Estimation Uncertainty of Lakes and Wetlands Using Satellites and Citizen Science**,** *IEEE JSTARS.* https://doi.org/10.1109/JSTARS.2023.3250354.
13. P. Das, F. Hossain, **S. Khan**, N. K. Biswas, H. Lee, T. Piman, C. Meechaiya, U. Ghimire, K. Hosen (2022) Reservoir Assessment Tool 2.0: Stakeholder driven Improvements to Satellite Remote Sensing based Reservoir Monitoring**,** *Environmental Modeling and Software.* https://doi.org/10.1016/j.envsoft.2022.105533.
14. **S. Khan,** N. Kamboj, U. Bhatia (2020) Lifeline Infrastructures and Hydro-climate Extremes: A Future Outlook, *Climate Change and Extreme Events*. https://doi.org/10.1016/B978-0-12-822700-8.00004-4.

Conference Presentations

1. **S. Khan,** F. Hossain, B. Davis, E. Roth, Mamadou Ba, M. Ahamed, and K. Islam (2025). sDRIPS: A Cloud-Based, Open-Source Python Package for Satellite- Informed Surface Water Irrigation Optimization. *AGU Conference 2025*.
2. **S. Khan,** F. Hossain, M. Ahamed (2024). sDRIPS: Satellite Data Rendered Irrigation Using Penman and SEBAL - A Framework For Surface Water Irrigation Optimization. *AGU Conference 2024.*
3. **S. Khan,** F. Hossain, N. Holterhoff, V. Kwok, A. Line, E. Roth, M. Ba. (2024). sDRIPS-Sense: A Hybrid Framework of Satellite and Sensor-Based Data Rendered Irrigation Using Penman and SEBAL Model. *AGU Conference 2024*.
4. T. Pavelsky, F. Hossain, S. K. Ghafoor, G. Parkins, A. M. Gomez, **S. Khan,** R. Bhattarai and M. Hendrickson (2024). Building a Global Lake Observation System through the Lake Observations by Citizen Scientists and Satellites (LOCSS) Project. *AGU Conference 2024.*
5. **S. Khan**, F. Hossain, T. Pavelsky, S. K. Ghafoor, G. Parkins, A. M. Gomez, S. Minocha, M.A. Bhuyan, T. A. Fayyaz, P.K Sarker, P.P Borua (2023). An Optimal Network Design Framework for Citizen Science-Satellite Monitoring of Surface Water Volume Changes in Bangladesh. *AGU Conference 2023.*
6. A. M. Gomez, T. Pavelsky, G. Parkins, M. Lane, **S. Khan**, F. Hossain, R. Bhattarai, S. K. Ghafoor, J.F. Crétaux (2023). Regional lake monitoring network design aided by Citizen Scientists and Satellites *AGU Conference 2023*.
7. S. Minocha, F. Hossain, P. Das, S. Suresh, **S. Khan**, G. Darkwah (2023). Collaborative Water Management for Advancing Open Science in Regulated River Basins with the Open-Source Reservoir Assessment Tool (RAT) 3.0: A Python Package Integrating Cloud Computing, Satellite Data, and Modeling *AGU Conference 2023*.
8. S. Suresh, F. Hossain, S. Minocha, P. Das, **S. Khan**, H. Lee, K. Andreadis, P. Oddo (2023). Satellite Earth Observations Based Tracking of Reservoir Operations for Flood Preparedness in Mountainous and High Precipitation Regions: A Case of the 2018 Kerala Floods. *AGU Conference 2023.*
9. A. M. Gomez, D.R. Arias, T. Pavelsky, G. Parkins, M. Lane, L.D. Donado, **S. Khan**, F. Hossain, W.J.G Ríos, R. Bhattarai, S.K. Ghafoor (2023). Enhancing levels of engagement in citizen science projects involving lake water level monitoring. *AGU Conference 2023*.
10. **S. Khan**, F. Hossain, T. Pavelsky, et.al (2022). Investigating Volumetric Uncertainty of Lakes and Wetlands Using Satellites and Citizen Science, *AWRA Conference 2022.*
11. A. M. Gomez, S. Biancamaria, T. Pavelsky, G. Parkins, M. Lane, **S. Khan**, F. Hossain, R. Bhattarai, S.K. Ghafoor, J.F. Crétaux, N. Picot (2022). Nadir altimeter validation in small lakes using multisource ground observations. *AGU Conference 2022.*
12. **S. Khan**, D. Upadhyay, U. Bhatia, Extreme Precipitation Volatilities and Its Implication for Critical Infrastructures in India (2020), *AMS Conference 2020.*

Magazine Articles

1. P. Das, S. Minocha, **S. Khan**, and F. Hossain. How satellites helped debunk 2024 flood myths. In *International Water Power and Dam Construction*, 2025. *Link*
2. F. Hossain, P. Das, G. Brencher, H. Conroy, G. Darkwah, A. McCall, S. Minocha, G. Schelepp, S. Yao, **S. Khan**, A satellite remote sensing perspective on water resources. In *International Water Power and Dam Construction*, 2023. *Link*
3. P. Das, F. Hossain, H. B. Helgason, and **S. Khan**. Satellites over the Amazon capture the choking of the 'house of god' by the Belo Monte dam – they can help find solutions, too. In *The Conversation*, 2022. *Link*

Textbooks

1. Contributed tutorials and exercises to the textbook - Hossain, F. (2025). Satellite Remote Sensing for Water Management, *Cambridge: Cambridge University Press*. https://doi.org/10.1017/9781009453509

Journal Peer Review Activity

Geophysical Research Letters (3)	Environmental Modeling and Software (1)	Earth and Space Science (1)
Remote Sensing of Environment (1)	Scientific Reports (1)	Earth Science Informatics (1)
Discover Geoscience (1)		

Research Experience

Graduate Research Assistant at the **University of Washington**, Seattle, WA *September 2021 – Present*

- **Irrigation advisories** – Developed sDRIPS, a satellite- and forecast-driven irrigation decision-support system for estimating net irrigation demand and generating weekly advisories for farmers and canal operators at the farm and command-area scale. Fused multi-source satellite imagery, including optical, thermal, and radar data, with numerical weather prediction outputs, precipitation nowcasts/forecasts, soil moisture, and field conditions to drive Penman-Monteith and SEBAL evapotranspiration models, percolation estimates, and field-scale water requirement calculations. Designed automated geospatial ETL workflows in Python and Google Earth Engine for large-scale data ingestion, preprocessing, harmonization, raster/vector processing, and product generation. Incorporated in situ sensors and local weather station data for statistical bias correction of coarse-scale meteorological and satellite-derived estimates, and characterized uncertainty using Generalized Likelihood Uncertainty Estimation (GLUE) and Monte Carlo analyses. Tool for public use (with link) - sDRIPS, sDRIPS-Sense, sDRIPS-TBP
- **Canal water allocation framework** – Developed a satellite-based canal water allocation methodology to help canal operators distribute water more efficiently based on crop water demand, actual field conditions, and canal network structure. Built the allocation framework on uncertainty-characterized net water requirement estimates from sDRIPS, ensuring distribution decisions account for input, parameter, and model uncertainty across diverse agro-climatic settings. Applied network-based analysis to translate field-scale irrigation demand into command-area and canal-level allocation decisions, supporting climate-change adaptation and more efficient agricultural water management. Tool for public use (with link) - sDRIPS-TBP
- **Scalable gauge network framework** – Designed an open-source framework integrating citizen science observations with current and future satellite missions to optimize gauge placement in lakes and wetlands, improving regional surface water storage monitoring in ungauged and data-sparse environments. Built Monte Carlo simulation workflows to quantify storage uncertainty and determine the optimal number of gauges needed for representative monitoring across heterogeneous hydrologic and landscape settings. Trained Random Forest models using hydrologic and landscape predictors, including precipitation, terrain metrics, waterbody area, and air temperature, to rank candidate sites and identify high-impact gauge locations. Framework on GitHub
- **Uncertainty in volumetric estimates** – Investigated uncertainty propagation in lake and wetland volume change estimates derived from satellite-based surface water extent and citizen science water-level observations. Built an ensemble of radar- and optical-based water classification approaches to improve robustness of surface water extent estimates, and evaluated sensitivity to sensor type, classification method, and spatial resolution, including comparison with high-resolution imagery such as PlanetScope at approximately 3 m resolution.

Summer Intern at **Jet Propulsion Laboratory**, **NASA**, Pasadena, CA *July 2025 – September 2025*

- Investigated PO.DAAC's SWOT On-Demand Raster Generation tool for producing customized raster products from SWOT satellite observations at user-defined spatial resolutions. Developed SWODLR-Python, a Python-based framework to streamline API-driven data requests, raster generation, subsetting, resampling, and processing. Evaluated customized SWOT raster products for hydrologic applications, including flood-propagation analysis in northeast Bangladesh. Framework on GitHub - SWODLR-Python

Undergraduate Research Work, **IIT-GN**, India: *May 2020 – Jan 2021*

- Analyzed precipitation-extreme volatility and infrastructure risk across India using observed rainfall data and CMIP5 Earth System Model outputs under RCP 8.5, including BNU-ESM, CanESM2, GFDL-ESM2M, MIROC-ESM, MPI-ESM-LR/MR, and NorESM.
- Applied Generalized Extreme Value analysis, 30-year and 100-year return levels, PEVI, Mann-Kendall trend testing, and Sen's slope to assess nonstationary hydroclimate extremes relevant to infrastructure design and climate-risk planning.
- Integrated hydroclimate-extreme analysis with network-science methods to evaluate lifeline infrastructure vulnerability and recovery strategies for Mumbai and Indian Railway networks.
- Published as a book chapter in Climate Change and Extreme Events: "Lifeline Infrastructures and Hydroclimate Extremes: A Future Outlook."

Teaching Experience

Teaching Assistant *2024, 2025*

- Taught Google Earth Engine (GEE) in *Satellite Remote Sensing for Water Resources* class at the *University of Washington,* with a focus on the Python API for quantifying satellite-based precipitation and evapotranspiration.

Invited Talks

- *CS4Env Seminar, University of Washington –* Delivered a lecture on "Satellite-based Irrigation around the World." *2026*
- *Environmental and Water Program Seminar, University of Washington's Civil and Environmental Engineering Department –* Delivered a lecture on utilizing satellite remote sensing to optimize surface water irrigation. *2024*
- Led tutorial class for *Satellite Remote Sensing for Water Resources* on estimating evapotranspiration through satellite data. *2023*

Mentor *2025*

- Supervised and taught the basics of geospatial data analysis and QGIS software to a high school student.

Trainer

- Trained 25 participants on '*Remote sensing based surface water tracking, citizen science, Cloud Computing and the Surface Water and Ocean Topography (SWOT) Mission*' (Dhaka, Bangladesh). *2023*
- Trained the engineers of *the Department of Agricultural Extension, Bangladesh,* on the use of the *Integrated Rice Advisory System (IRAS)* for irrigation advisory dissemination. *2023*
- Trainer at *3rd SWOT Early Adopter Virtual Hackathon*, organized by *NASA, CNES, and the University of Washington.* *2022*
- Trained engineers from *the Department of Hydrology and Meteorology, Nepal,* on *Utilizing Satellite Remote Sensing Data to Monitor Lakes*. *2022*

Leadership & Engagement

American Water Resources Association (AWRA), University of Washington Chapter *2023*

- Revived the student chapter post-COVID-19, fostering community and engagement in water resources.

Class Representative, Indian Institute of Technology Gandhinagar *2019, 2020*

- Represented peers in academic and administrative discussions, serving as the primary liaison between students and faculty

Stakeholders Engagement

Collaborated with government agencies and private sector partners to co-develop science-informed decision-support tools. Engagement activities included code sharing, technical feedback, and capacity-building training sessions to enhance operational adoption of research outputs.

- **Bangladesh Water Development Board** (A government agency that is responsible for water management in Bangladesh). *January 2022 - Present*

- **Department of Agricultural Extension** (A government agency in Bangladesh that provides extension services to the farmers in accessing and utilizing better know-how to increase profitable crop production). *January 2023 – Present*
- **Crown Farms** (A private farm in South Africa) *April 2024 – September 2025*
- **University of Washington Landscape Workers** *January 2024 - March 2025*
- **Pakistan Council of Research in Water Resources** (A government agency that is responsible for water management in Pakistan). *January 2022 - April 2022*

Other Tools Developed

1. SWOT Raster Visualizer – A visualization tool for SWOT raster products to monitor water surface elevation. Applied during the 2024 Tripura (India) floods.
2. Personal Website Chatbot – Developed a retrieval-augmented generation chatbot for AI-assisted information access, with a broader vision for adapting similar interfaces to Earth science decision-support systems.
3. IRAS – Provides weekly irrigation advisories for rice cultivation in Northeastern and Northwestern Bangladesh. The advisories are disseminated by the Department of Agricultural Extension to hundreds of thousands of farmers, supporting large-scale water management and food security.
4. PyWRIS – A Python API for streamlined data access to India's Water Resources Information System (WRIS).
5. Barind, Haors – Satellite based surface water availability in the Northeastern and Northwestern parts of Bangladesh respectively.
6. A detailed document on ReadtheDocs – Documentation for a satellite-driven irrigation advisory system, improving usability, reproducibility, and stakeholder adoption.

Media Highlights

- sDRIPS-Sense work featured in *University of Washington News*, News Link *2024*
- IRAS system featured in *NASA* article, News Link *2023*

Skills & Interests

- **Earth Observation, Remote Sensing, and Multimodal Data Integration:** Satellite remote sensing; optical, thermal, radar, microwave, and altimetry datasets; Landsat, MODIS, Sentinel-1 SAR, Sentinel-2, SMAP, SWOT, and PlanetScope; precipitation and hydrometeorological datasets including IMERG, CHIRPS, GSMaP, and GLDAS; reanalysis and forecast products including ERA5 and GFS; DEMs; in situ observations; CMIP climate simulations and global climate model outputs for hydroclimatic extremes, infrastructure risk, and climate-impact assessment, including CanESM2, GFDL-ESM2M, MIROC-ESM, MPI-ESM, and NorESM; multi-sensor, multi-resolution, and spatio-temporal data fusion.
- **Hydrology, Climate, and Agricultural Water Modeling:** Hydrologic and Earth system process modeling; hydroclimatic analysis; agricultural water management; irrigation demand estimation; evapotranspiration modeling; soil moisture estimation; percolation; water-balance modeling; drought resilience; surface water monitoring; reservoir, lake, and wetland storage estimation; climate adaptation; food-water systems; predictive systems for environmental processes.
- **Machine Learning, Statistics, and Uncertainty Quantification:** Statistical modeling; machine learning; Random Forests; regression and tree-based models; time series analysis; feature engineering; bias correction; ensemble modeling; Monte Carlo simulation; Generalized Likelihood Uncertainty Estimation; sensitivity analysis; uncertainty propagation; Generalized Extreme Value analysis; physics-informed and hybrid ML workflows; predictive modeling for environmental systems; working knowledge of deep learning for satellite image analysis.
- **Geospatial Analytics, Scientific Computing, and Data Engineering:** Python, Google Earth Engine Python and JavaScript APIs, Xarray, Dask, NumPy, Pandas, GeoPandas, Rasterio, SciPy, PyTorch, Scikit-learn, GDAL, ArcGIS, QGIS, MATLAB, Bash, Git, Linux/HPC environments, multiprocessing, parallel and distributed processing, scalable geospatial ETL, raster and vector analytics, spatial statistics, API-driven data ingestion, automated environmental data pipelines, and cloud/HPC-style large-scale analytics.

- **AI Systems, RAG, and Computer Vision:** Retrieval-augmented generation, LLM-assisted information systems, embedding-based retrieval, semantic search, document indexing and querying pipelines, chatbot development for research and decision-support systems, CNNs, U-Net, satellite image segmentation, and flood extent mapping.
- **Data Formats, Software Development, and Reproducibility:** NetCDF, HDF5, GeoTIFF, Cloud-Optimized GeoTIFF, metadata-aware processing, reproducible workflows, scientific package development, PyPI package publication, Read the Docs documentation, REST APIs, dashboards, Plotly, Matplotlib, technical documentation, and reproducible scientific workflows.
- **Decision Support, Deployment, and Stakeholder Engagement:** Operational irrigation advisories, agricultural decision-support systems, drought and water-risk indicators, scalable pipeline architecture, stakeholder-oriented tool development, government and private-sector engagement, capacity-building trainings, interdisciplinary collaboration, science communication, stakeholder reporting, proposal development, and leadership in translating computational research into operational environmental intelligence.
- **Society Affiliations**: Student Member at the American Geophysical Union (AGU) and the American Water Resources Association (AWRA).

References Available Upon Request

Dr. Faisal Hossain (fhossain@uw.edu) - University of Washington

Dr. Tamlin Pavelsky (pavelsky@email.unc.edu) - University of North Carolina

Edward Armstrong (edward.m.armstrong@jpl.nasa.gov) - Jet Propulsion Laboratory, NASA

Karen Yuen (karen.yuen@jpl.nasa.gov) - Jet Propulsion Laboratory, NASA